

Surface-Consistent Multiples: Mathematical Explanation

Converted from Lab1_calculations.m

Overview

This document explains the nested loops that enumerate surface-consistent multiples and place impulses at their total two-way travel times. Indices n, m, p refer to reflecting interfaces as used in the MATLAB code.

Index Roles and Reflection Signs

- n : index of the first upgoing reflection (top-up)
- m : index of the downgoing reflection (iterates from $n - 1$ down to 0)
- p : index of the final upgoing reflection (iterates from $m + 1$ to end).

Reflection coefficients:

$$R_1 = R(n), \quad R_3 = R(p),$$

and the downgoing reflection is

$$R_2 = \begin{cases} -R(m), & m > 0, \\ -1, & m = 0 \text{ (surface reflection)}. \end{cases}$$

Segment Definitions

For a chosen triple (n, m, p) define

$$\text{high} = \min(n, p), \quad \text{low} = \max(n, p).$$

Three path segments are considered: - Upper segment (above the higher interface): layer indices $\mathcal{A} = \{1, \dots, \text{high}\}$. - Lower segment (between higher and lower interfaces): $\mathcal{B} = \{\text{high} + 1, \dots, \text{low}\}$. - Middle segment (between downgoing reflection and higher interface): $\mathcal{C} = \{m + 1, \dots, \text{high}\}$.

Layerwise two-way travel times: for layer index i ,

$$\Delta t_i = \frac{2 \Delta z_i}{v_p(i)}.$$

Segment two-way times are sums:

$$\text{tw}_{\mathcal{A}} = \sum_{i \in \mathcal{A}} \Delta t_i, \quad \text{tw}_{\mathcal{B}} = \sum_{i \in \mathcal{B}} \Delta t_i, \quad \text{tw}_{\mathcal{C}} = \sum_{i \in \mathcal{C}} \Delta t_i.$$

Transmission Factors

Transmission factors use the interface reflection coefficients $R(i)$. For a contiguous index set $\mathcal{I}$ with elements in ascending order, transmissions are computed by excluding the layer where the reflection itself occurs (code uses ‘indaT’, ‘indbT’, ‘indcT’). Symbolically:

$$T(\mathcal{I}) = \prod_{j \in \mathcal{I}_T} (1 - R(j)^2),$$

where $\mathcal{I}_T$ denotes the subset of $\mathcal{I}$ excluding the last index corresponding to the reflection in that segment. In the code:

$$T_a = \prod_{j \in \text{indaT}} (1 - R(j)^2), \quad T_b = \prod_{j \in \text{indbT}} (1 - R(j)^2), \quad T_c = \prod_{j \in \text{indcT}} (1 - R(j)^2).$$

Attenuation Factors

Per-layer attenuation uses quality factor $Q(i)$ and the two-way time:

$$A_i = \exp\left(-\frac{32 \Delta t_i}{2 Q(i)}\right).$$

Segment attenuation products:

$$Q_a = \prod_{i \in \mathcal{A}} A_i, \quad Q_b = \prod_{i \in \mathcal{B}} A_i, \quad Q_c = \prod_{i \in \mathcal{C}} A_i.$$

Arrival Time and Amplitude

The total arrival two-way time for the multiple formed by (n, m, p) is

$$t_{\text{mult}} = \text{tw}_{\mathcal{A}} + \text{tw}_{\mathcal{B}} + \text{tw}_{\mathcal{C}}.$$

In discrete time this maps to the nearest sample t_{ind} where $t(t_{\text{ind}}) \approx t_{\text{mult}}$.

The amplitude without attenuation (but including transmission factors) is

$$A_{\text{no-atten}} = R_1 R_2 R_3 T_a T_b T_c.$$

Including attenuation:

$$A_{\text{atten}} = R_1 R_2 R_3 T_a T_b T_c Q_a Q_b Q_c.$$

The code accumulates these into the multiple series:

$$M_1(t_{\text{ind}}) += A_{\text{no-atten}}, \quad M_2(t_{\text{ind}}) += A_{\text{atten}}.$$

Algorithmic Mapping

The triple nested loops iterate:

$$\text{for } n = 1 : \dots, \text{ for } m = n - 1 : \dots, 0, \text{ for } p = m + 1 : \dots$$

Each iteration:

1. Sets R_1, R_2, R_3 .
2. Determines $\mathcal{A}, \mathcal{B}, \mathcal{C}$ and computes T_a, T_b, T_c , tw_* , and Q_* .
3. Computes t_{mult} and finds the nearest time index.
4. Adds amplitudes to M_1 and M_2 .

Notes

- The sign convention for R_2 accounts for downgoing reflection or a surface reflection. - Transmission products skip the reflecting interface layer where appropriate, matching the code's sliced index sets.